

A
PROJECT REPORT ON
“Project Title”
SUBMITTED TO
D. Y. PATIL UNIVERSITY

**A Project report submitted towards the partial fulfillment of the
requirements for the Degree of
BACHELOR OF TECHNOLOGY IN <DEPARTMENT NAME>**

SUBMITTED BY
Name of Student (Exam Seat No./PRN No)
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UNDER THE GUIDANCE OF
Name of Guide



DEPARTMENT OF <DEPARTMENT NAME>
SCHOOL OF ENGINEERING AND TECHNOLOGY
D. Y. PATIL UNIVERSITY

A/P Ambi, MIDC Road, Talegaon Dabhade, PUNE-410507

Academic Year 2025-2026

DEPARTMENT OF INFORMATION TECHNOLOGY
SCHOOL OF ENGINEERING AND TECHNOLOGY
D. Y. PATIL UNIVERSITY

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CERTIFICATE

This is certified that the project entitled

“Project Title”

Submitted By

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Is a record of bonafide work carried out by them, in the partial fulfilment of the requirement for the award of Degree of Bachelor of Technology (Department Name) at School Of Engineering And Technology, D. Y. Patil University Pune. This work is done during year 2025- 2026, under our guidance.

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Acknowledgements

We are profoundly grateful to <Guide Name> for his expert guidance and continuous encouragement throughout to see that this project rights its target since its commencement to its completion.

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ABSTRACT

Keywords:

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CHAPTER 1

INTRODUCTION

1.1 INTRODUCTION

1.1.1 Problem Identified

1.1.2 Problem Statements

- **Statement of scope:**

1.1.3 Motivation and Need

1.1.4 Project Idea

CHAPTER 2

LITERATURE SURVEY

2.1 STUDY OF RESEARCH PAPER

1. Paper Name:

Author:

Description :

2. Paper Name:

Author:

Description :

3. Paper Name:

Author:

Description:

4. Paper Name:

Author:

Description:

5. **Paper Name:**

Author:

Description:

6. **Paper Name:**

Author:

Description:

7. **Paper Name:**

Author:

Description:

8. **Paper Name:**

Author:

Description:

Sr. no	Paper Name	Author Name	Description	Remark
1.	Novel Transfer Learning Approach for Driver Drowsiness Detection Using Eye Movement Behavior	Hamza Ahmad Madni, Ali Raza Rukhshanda Sehar, Nisrean Thalji, Laith Abualigah	Drivers drowsiness and fatigue decreases vehicle management skills of a driver. Driver in a drowsiness state is the one among the important reason of increasing amount of road accidents and death. Hence the drowsiness detection of driver is considering as most active research field	The VGLG approach combines VGG-16 for feature extraction and LightGBM for fast detection, achieving 99% accuracy in 0.00829 seconds, offering potential to prevent drowsiness-related accidents.
2.				
3.				
4.				
5.				
6.				
7.				
8.				

CHAPTER 3

PROBLEM DEFINITION

3.0.1 Proposed Problem Definition

3.0.2 Modules

3.0.3 Existing System

- **Disadvantages of Existing System:**

3.0.4 Area of Project

3.0.5 Project Scope

3.0.6 DETAILS OF ALGORITHMS

3.0.7 Goal

3.0.8 Objective

CHAPTER 4

HARDWARE AND SOFTWARE REQUIREMENTS

4.1 EXTERNAL INTERFACE REQUIREMENT

4.1.1 User Interface

4.1.2 Hardware Interfaces:

Major Constraints

- Hardware :
- RAM : 8GB
- Key Board: Standard Windows Keyboard

The camera catches the picture and sends the processor of the PC which comprises of a 32-bit memory card introduced with OpenCV which helps in picture handling. On the off chance that the sign passes the boundary of a bunch of persistent casings with EAR not as much as limit esteem, it will naturally make the alert blare and the speed of the vehicle gets diminished. In any case that sign is dismissed and the following sign is handled.

4.1.3 Software Requirements

- Operating System: Windows 7 Or windows 10
- IDE: Anaconda ,spyder
- Programming Language : Python

4.1.4 Software requirement Specification Purpose and Scope of Document

4.2 NON FUNCTIONAL REQUIREMENT

4.2.1 Performance Requirements

4.2.2 Safety Requirement

4.2.3 Software Quality Attributes

- Maintainability: After the deployment of the project if any error occurs then it can be easily maintained by the software developer.
- Reliability: The performance of the software is better which will increase the reliability of the Software.
- User Friendliness: Since, the software is a GUI application; the output generated is much user friendly in its behavior.
- Integrity: Integrity refers to the extent to which access to software or data by unauthorized persons can be controlled.
- Security: Users are authenticated using many security phases so reliable security is provided.
- Testability: The software will be tested considering all the aspects.

CHAPTER 5

SYSTEM DESIGN

5.1 SYSTEM ARCHITECTURE

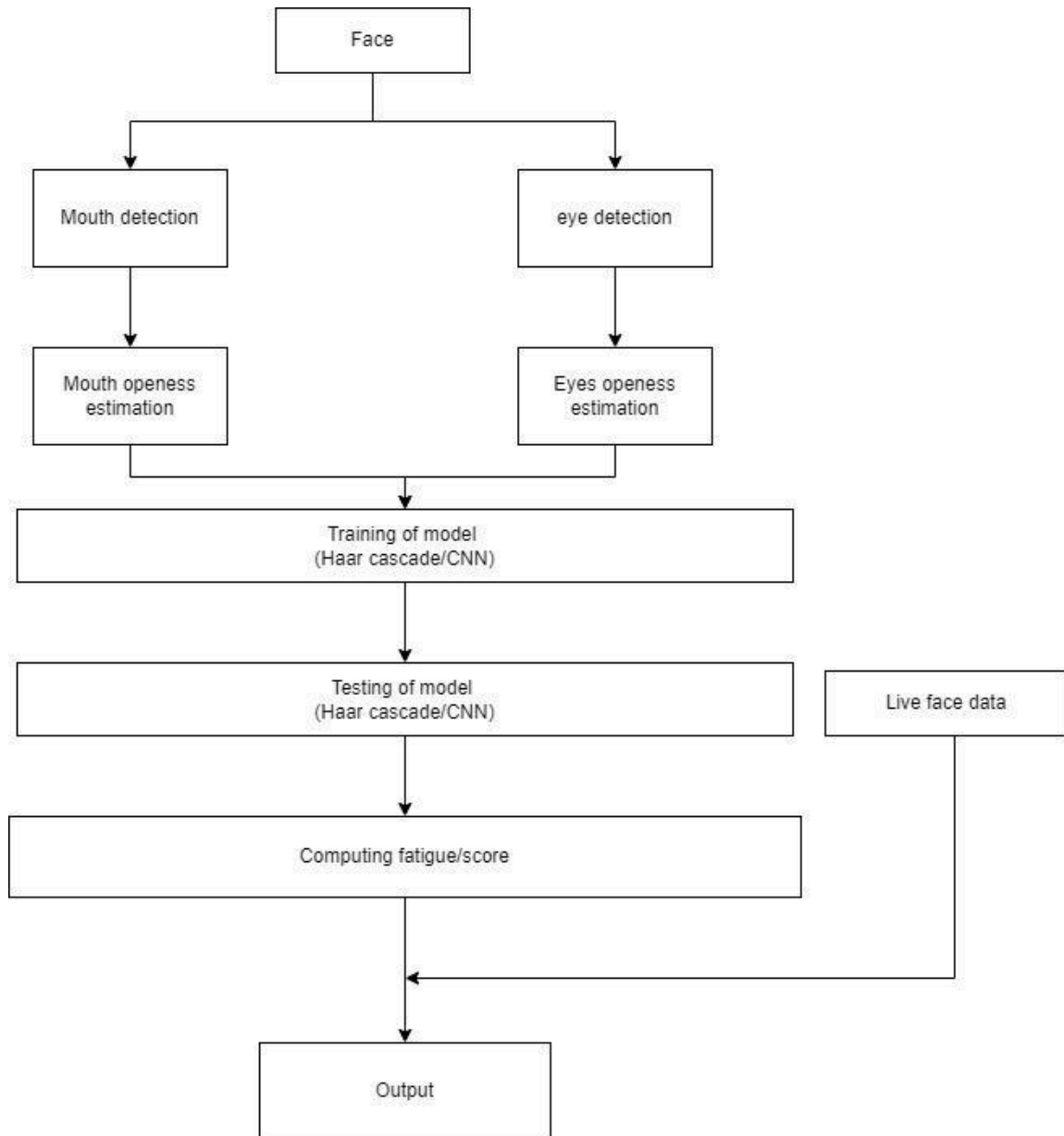


Figure 5.1: System architecture

5.2 DATA FLOW DIAGRAM

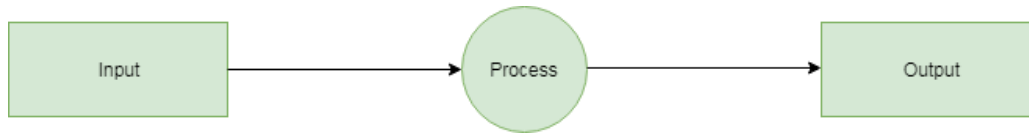


Figure 5.2.1: DFD 0

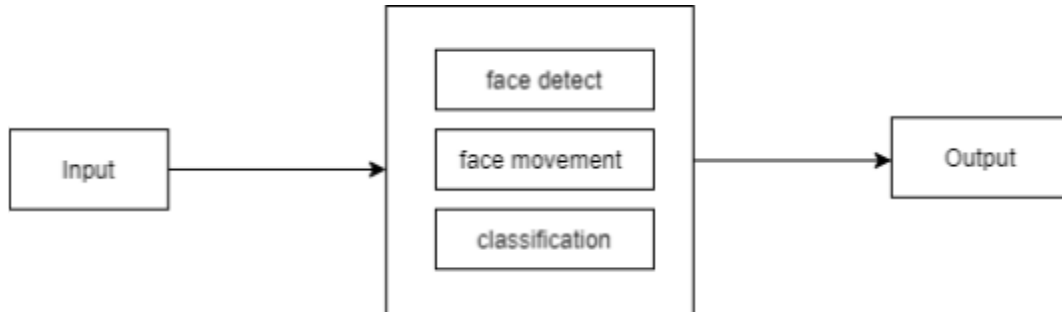


Figure 5.2.2: DFD 1

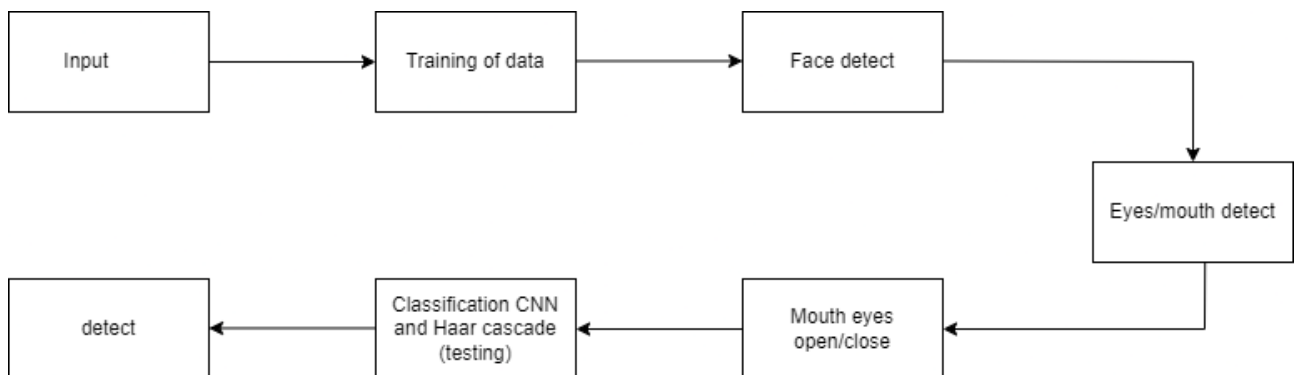


Figure 5.2.3: DFD 2

5.3 UML DIAGRAM

Following diagram shows the UML Diagram of this system.

1. Usecase Diagram
2. Activity Diagram
3. Sequence Diagram
4. Class Diagram

5.4 USE CASE DIAGRAM

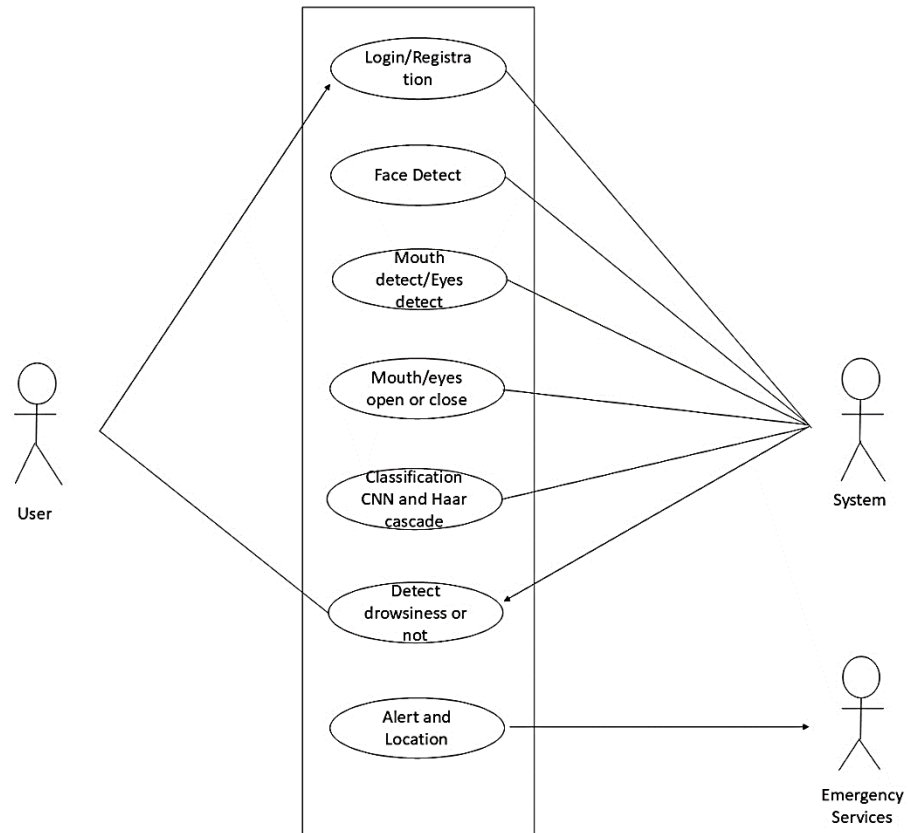


Figure 5.4: Use Case Diagram

5.5 ACTIVITY DIAGRAM

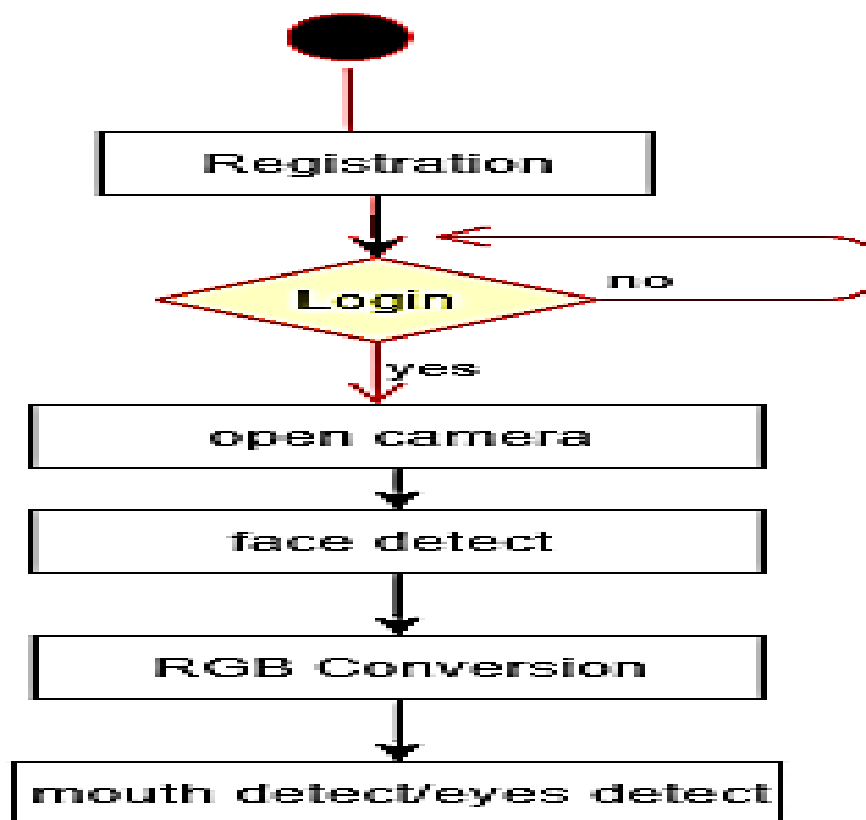


Figure 5.5: Activity Diagram

5.6 SEQUENCE DIAGRAM

a) Training Diagram

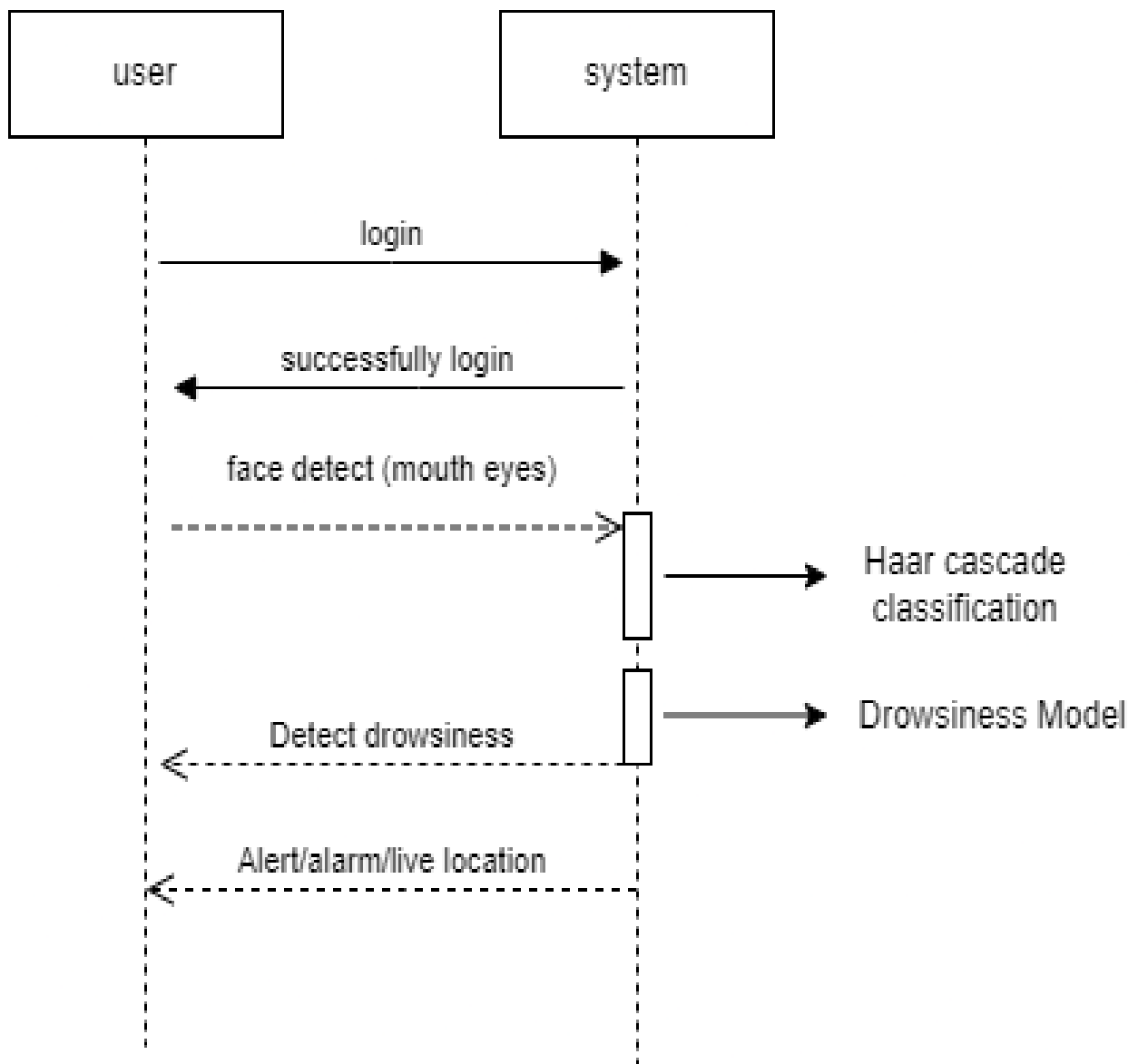


Figure 5.6.1: Training Sequence Diagram

b) Testing Diagram

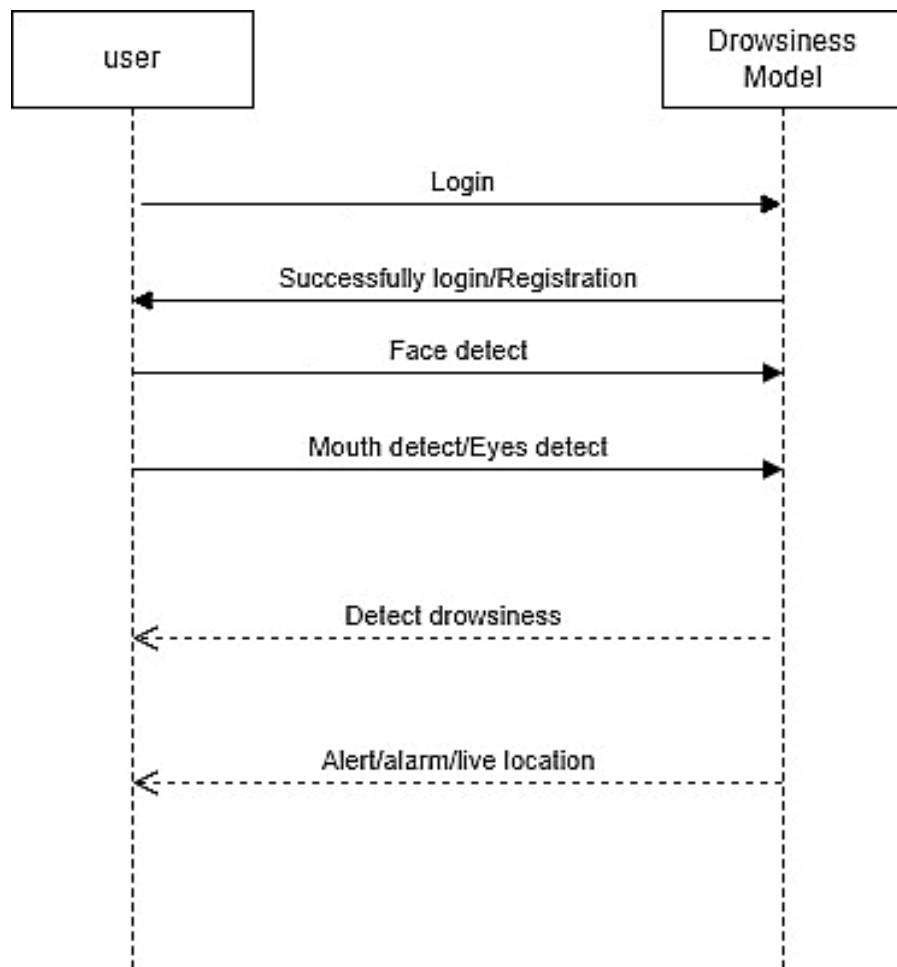


Figure 5.6.2: Testing Sequence Diagram

5.7 CLASS DIAGRAM

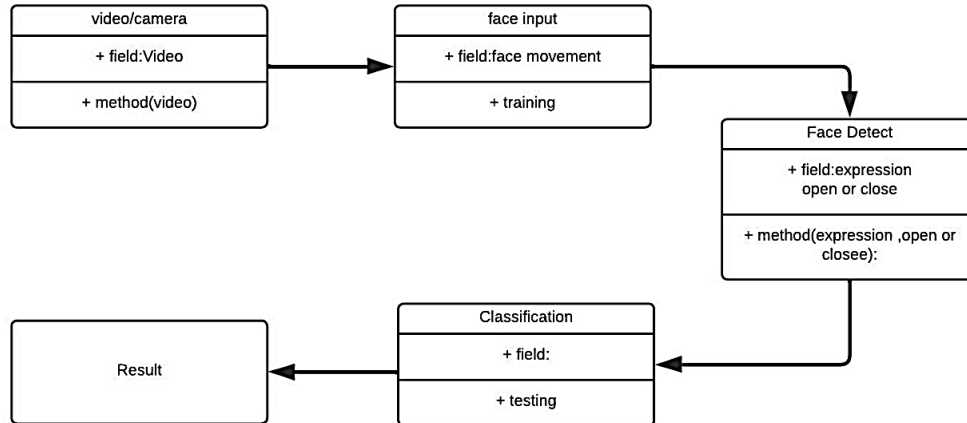


Figure 5.7: Class Diagram

5.8 FLOWCHART DIAGRAM

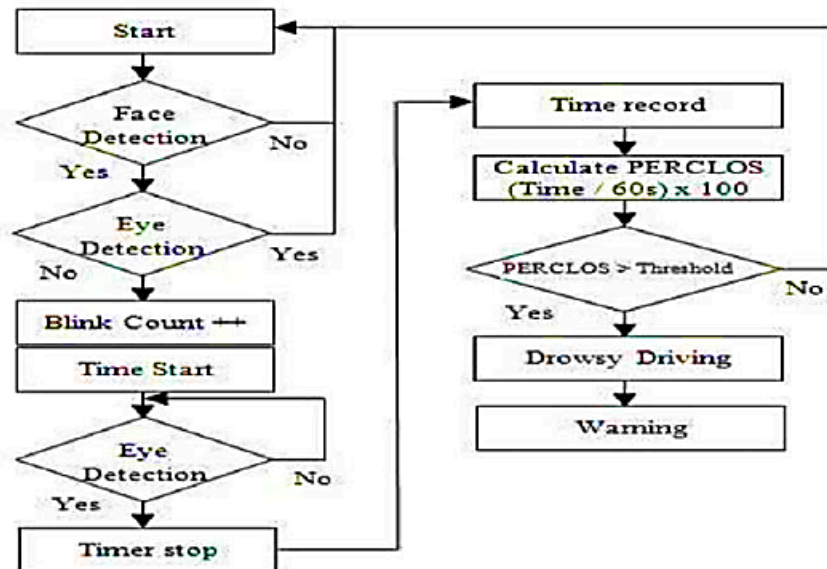


Figure 5.8: Flowchart Diagram

5.9 MATHEMATICAL MODEL

Let S is the Whole System Consist of

$S = \{I, P, O\}$

Where,

$I = \{CUR_LOC, SE_LOC, LOG, RE, PRO\}$

LOG=user login into system

SE_P= Select Image

PRO= Data set

TI = Data Pre-processing

P = Process

- **Steps**

Step1: User will login

Step2: User will select Image

Step3: User will Analyze Data

Step3: User will Apply CNN Algorithm

Step4: System will give Prediction the Driver Drowsiness or Not

OUTPUT: display result on system

CHAPTER 6

SOFTWARE INFORMATION

6.1 PYTHON:

What is Python?

Python is a high-level, interpreted, interactive and object-oriented scripting language. Python is designed to be highly readable. It uses English keywords frequently where as other languages use punctuation, and it has fewer syntactical constructions than other language. Python is Interpreted- Python is processed at runtime by the interpreter. You do not need to compile your program before executing it. This is similar to PERL and PHP. Python is Interactive You can actually sit at a Python prompt and interact with the interpreter directly to write your programs. Python is Object-Oriented Python supports Object-Oriented style or technique of program- ming that encapsulates code within objects. Python is a Beginner's Language Python is a great language for the beginner-level programmers and supports the development of a wide range of applications from simple text processing to WWW browsers to games.

Tools Used Are:

OpenCV python OpenCV-Python is a library of Python bindings designed to solve computer vision problems. OpenCV (Open Source Computer Vision Library) is an open-source BSD-licensed library that includes several hundreds of computer vision algorithms. OpenCV-Python is the Python API for OpenCV, combining the best qualities of the OpenCV C++ API and the Python language. OpenCV-Python makes use of Numpy, which is a highly optimized library for numerical operations with a Python-style syntax. All the OpenCV array structures are converted to and from Numpy arrays. This also makes it easier to integrate with other libraries that use Numpy such as SciPy and Matplotlib.

1. **TensorFlow:** TensorFlow is an open-source software library for machine learning across a range of tasks. It is a symbolic math library, and also used as a system for building and training neural networks to detect and decipher patterns and correlations, analogous to human learning and reasoning. Tensor-Flow provides a Python API as well as C++,Haskell,Java,Go and Rust API

2. **Anaconda:** Anaconda is a free and open-source distribution of the Python and R programming languages for scientific computing (data science, machine learning applications, large-scale data processing, predictive analytics, etc.), that aims to simplify package management and deployment. Package versions are managed by the package management system conda. The Anaconda distribution is used by over 13 million users and includes more than 1400 popular data-science packages suitable for Windows, Linux, and MacOS. Anaconda distribution comes with more than 1,400 packages as well as the Conda package and virtual environment manager, called Anaconda Navigator, so it eliminates the need to learn to install each library independently. The open-source packages can be individually installed from the Anaconda repository with the conda install command or using the pip install command that is installed with Anaconda. Pip packages provide many of the features of conda packages and in most cases they can work together. Custom packages can be made using the conda build command, and can be shared with others by uploading them to Anaconda Cloud, PyPI or other repositories.
3. **Anaconda Navigator:** Anaconda Navigator is a desktop graphical user interface (GUI) included in Anaconda distribution that allows users to launch applications and manage conda packages, environments and channels without using command-line commands. Navigator can search for packages on Anaconda Cloud or in a local Anaconda Repository, install them in an environment, run the packages and update them. It is available for Windows, macOS and Linux. The following applications are available by default in Navigator:

We are using spyder for execution of developed system

4. **Spyder:** Spyder is an open-source cross-platform integrated development environment (IDE) for scientific programming in the Python language. Spyder integrates with a number of prominent packages in the scientific Python stack, including NumPy, SciPy, Matplotlib, pandas, IPython, SymPy and Cython, as well as other open-source software. It is released under the MIT license.

Features:

- (a) An editor with syntax highlighting, introspection, code completion.
- (b) Support for multiple IPython consoles.
- (c) The ability to explore and edit variables from a GUI.
- (d) A Help pane able to retrieve and render rich text documentation on functions, classes and methods automatically or on-demand.
- (e) A debugger linked to IPdb, for step-by-step execution.
- (f) A run-time explorer, to benchmark code Project support, allowing work on multiple development efforts simultaneously. 7. A built-in explorer, for interacting with the system and managing projects A “Find in Files” feature, allowing full regular expression search over a scope.

CHAPTER 7

PROJECT PLAN

7.1 PROJECT SCHEDULING

Timeline Chart:

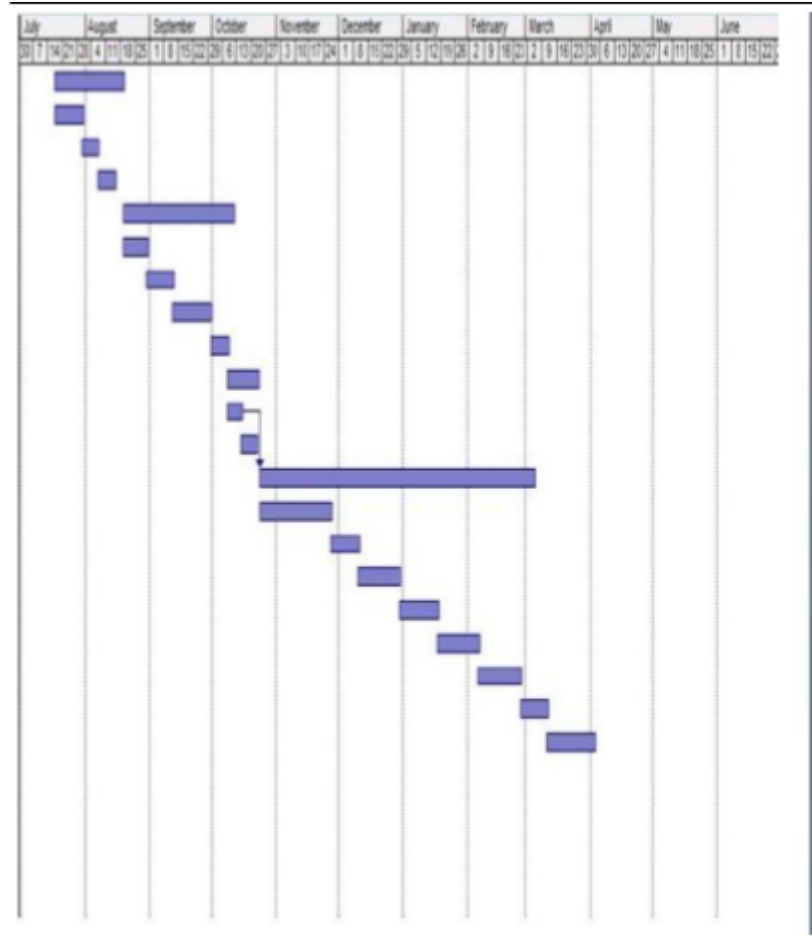


Figure 7.1: Timeline Chart

Team Organization: Team consists of 4 members and proper planning mechanism are used and roles of each member are defined.

Team structure: The team structure for the project is identified. There are total 4 members in our team and roles are defined. All members are contributing in all the phases of project.

Month Scheduled	Phase	Work Done
June-August	Topic Searching	Topic Searched
August-September	Topic Selection	Topic Selected
August-September	Project Confirmation	Project Confirmed
August-September	Literature survey	Literature survey done
Sept-Oct	Requirement Analysis	Requirement Analysis done
Sept-Oct	Requirement Gathering	Requirement Gathered
Nov-Dec	Designing	Architecture Design
Nov-Dec	Designing Test	GUI Tested
Nov-Dec	Database Creation	Database Tested
Jan-Feb	Coding	Coded Different Modules
Jan-Feb	Database and modules connectivity	Connectivity done
Mar	Testing of project	Project Tested
April	Result Analysis	Result Analysis

Management reporting and communication: Well planning mechanisms are used for progress reporting and inter/intra team communication are identified as per requirements of the project.

7.2 PROJECT PLAN

N O	TASK	DURATI ON(Days)	S T A R T	D A T E	E N D	D A T E
1	Group Formation	4	-	-		
2	Decide Area Of Interest	4	-	-		
3	Search topic	5	-	-		
4	Topic Selection	5	-	-		
5	Section Topic	5	-	-		
6	Search-Related Information	12	-	-		
7	Understanding concept	7	-	-		
8	Search essential Document (IEEE and White paper, software)	6	-	-		
9	Problem Definition	2	-	-		
10	Literature	5	-	-		

	Survey			
11	SRS	14	-	-
12	Project planning	2	-	-
13	Modeling and design	10	-	-
14	Technical Specification	2	-	-
15	Implementatio n & Testing	6	-	-

7.3 GANTT LIST

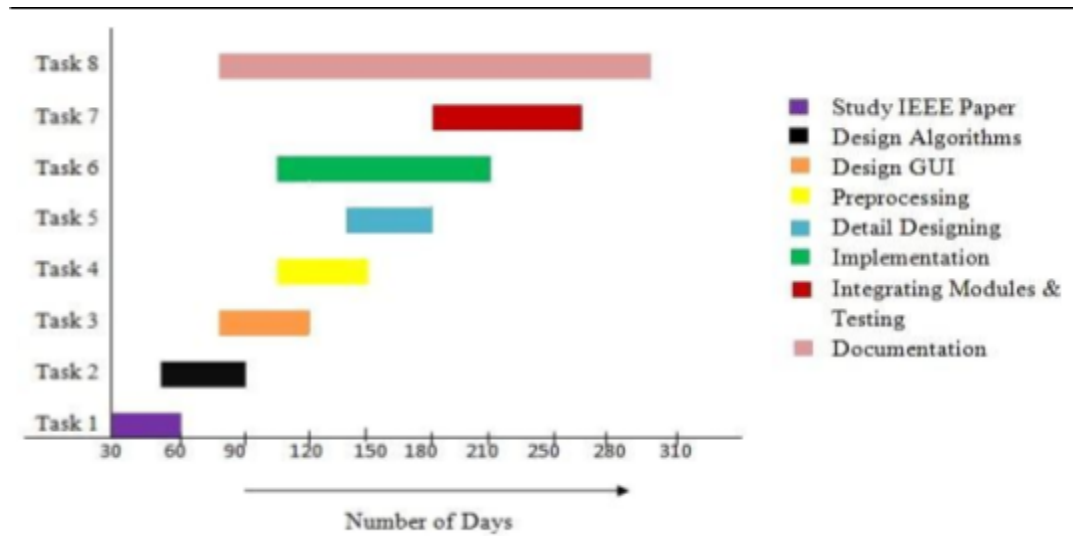


Figure 7.2: Gantt Chart

7.4 RESPONSIBILITIES OF DEVELOPER

1. Working with the Project Manager on definition of development requirements and priorities.
2. Data Migration.
3. Interfaces with other systems.
4. Reporting configuration and deployment.
5. Set up and maintenance of security rights and access permissions.
6. Contributing to technical strategy, policy and procedure.
7. Development and operation of technical testing programs.
8. Production of technical documentation to agreed quality standards.

9. Reporting on progress/issues to management and users.

7.4.1 Overview of responsibilities of Developer

1. To have understanding of the problem statement.
2. To know what are the hardware and software requirements of proposed system.
3. To have understanding of proposed system.
4. To do planning various activities with the help of planner.
5. Designing, programming, testing etc.

7.5 Test Cases

CHAPTER 8

OTHER SPECIFICATION

8.1 ADVANTAGES OF SYSTEM

8.2 APPLICATIONS

CHAPTER 9

RESULTS AND DISCUSSION

9.1 RESULTS

9.1.1 DISCUSSION

CHAPTER 10

CONCLUSION

10.1 CONCLUSION

10.2 FUTURE SCOPE

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